E-Commerce Customer Segmentation

RFM analysis | Veda Technology internship | Evidence-based recommendations

4,338 customers | **392,692 cleaned lines** | **2 data-supported clusters** | **Reference date: 2011-12-10**

Segment	Customers	Avg recency	Avg invoices	Avg spend	Revenue share
Champions	1,668	26.5 d	8.44	4,535.74	85.1%
Lost / Low-Engagement Customers	2,670	134.7 d	1.67	494.98	14.9%

Who to target first

Protect the Champions group first. It contains 1,668 customers and contributes 85.1% of cleaned historical revenue (about £7.57m in the dataset currency). Their mean recency is only 26.5 days; rewarding repeat behavior helps defend the strongest demonstrated value. Use exclusive early access, VIP recognition, bundles, and loyalty benefits while measuring margin and repeat purchase.

How to win back lower-engagement customers

The Lost / Low-Engagement segment is larger (2,670 customers) but averages 134.7 days since purchase, 1.67 invoices, and £495 historical spend; it contributes 14.9% of revenue. Run a limited, personalized reactivation test (relevant product recommendations and a time-limited offer), with a randomized holdout. Suppress non-responders from repeated discounting and evaluate incremental gross margin, not just clicks or sales.

Retention and measurement

For valuable customers: monitor recency drift, send useful replenishment/product reminders, offer tiered loyalty rewards, and prioritize service recovery. For win-back tests, compare conversion, repeat purchase and contribution margin against a holdout; cap discount cost. Use country and product preferences to personalize communications where consent and applicable privacy rules permit.

How segments were derived

After removing 5,268 exact duplicates, 135,037 rows without CustomerID, 8,872 cancelled-invoice rows, and 40 rows with non-positive quantity/price, 392,692 transaction lines remained. RFM features were log-transformed and standardized before K-Means. k=2 had the strongest sampled silhouette (0.433) among k=2–8; the elbow chart is included. The analysis supports two broad groups, not a fabricated four-way taxonomy.

Limitations

The workbook covers Dec 2010–Dec 2011; these are historical patterns, not current customer status. Monetary represents item revenue, not profit, and no campaign response, cost or customer demographics are provided. Cluster labels summarize relative behavior and are not causal or ground-truth classifications. Validate any campaign with current data and controlled tests.